

DS&AI

Machine Learning

DPP: 1

k-nearest neighbor

Q1 Consider a set of five training examples given as $((x_i, y_i), c_i)$ values, where x_i and y_i are the two attribute values (positive integers) and c_i is the binary class label:

- $((1, 1), -1)$
- $((1, 7), +1)$
- $((3, 3), +1)$
- $((5, 4), -1)$
- $((2, 5), -1)$

Classify a test example at coordinates (3, 6) using a k-NN classifier with $k = 3$ and Manhattan distance defined by $d((u, v), (p, q)) = |u-p| + |v-q|$

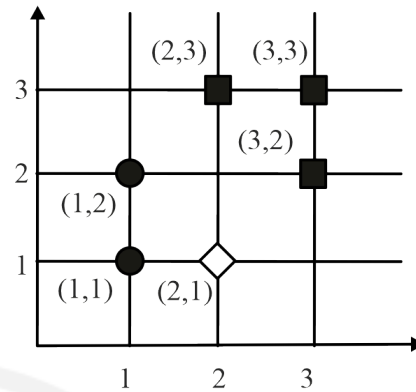
Q2 Predicting Movie Genre

IMDB Rating	Duration	Genre
8.0 (Mission Impossible)	160	Action
6.2(Gadar 2)	170	Action
7.2 (Rocky & Rani)	168	Comedy
8.2 (OMG 2)	155	Comedy

The genre of "Barbie" movie with IMDb rating 7.4 and duration 114.

Now predict the genre of.....

Q3 Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in nearest neighbor algorithm for which the diamond(♦) shaped data point is assigned the label square is



Q4 Which of the following are true for the k-nearest neighbour (k-NN) algorithm?

- (A) K-NN can be used for both classification and regression.
- (B) As k increases, the bias usually increases.
- (C) The decision boundary looks smoother with smaller values of k .
- (D) As k increases, the variance usually increases.

Q5

Sl. No.	Swim	Fly	Crawl	Class
1	Fast	No	No	Fish
2	Fast	No	Yes	Animal
3	Slow	No	No	Animal
4	No	No	No	Animal
5	No	Short	No	Bird
6	No	Short	No	Bird
7	No	Rarely	No	Animal
8	Slow	No	Yes	Animal
9	Slow	No	No	Fish
10	Slow	No	Yes	Fish
11	No	Long	No	Bird
12	Fast	No	No	Bird

The test instance (Slow, Rarely, No)

Q6 In k-NN, what is the effect of using weighted voting for classification?

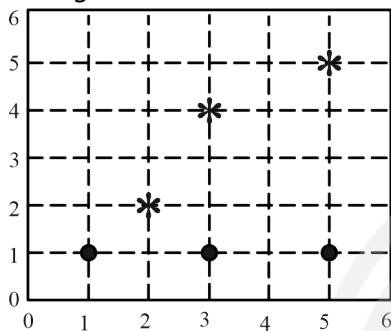
- (A) It gives equal importance to all neighbors
- (B)


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It assigns more importance to closer neighbors

- (C) It reduces the effect of outliers
(D) It increases the model's bias

- Q7** What is the training error (fraction labeled wrong) in the above picture using 1-nearest neighbor and the L_1 distance norm between points? If there is more than one equally close nearest neighbor, use the majority label of all the equally close nearest neighbors. Please make sure you really are computing training error, not testing error.



- (A) 0
(B) 1/6
(C) 2/6
(D) 4/6

- Q8** Consider a 2D dataset where training points are on the integer grid (x_1, x_2) , and both dimensions x_1 and x_2 range from 1 to 100 (inclusive). The binary label for point (x_1, x_2) is $(-1)^{(x_1+x_2)}$. What is the label for test point $(51.3, 62.1)$ with a 3-NN classifier?

- Q9** Which of the following statements is/are true in case of k-NN?
- (A) In case of very large value of k , we may include points from other classes into the neighborhood.
(B) In case of too small value of k , the algorithm is very sensitive to noise.
(C) K-NN is non-parametric algorithm
(D) k-NN algorithm does more computation on test time rather than train time.

- Q10** Consider the following dataset for a 3-nearest neighbor (3NN) classification problem:

Point	Coordinates	Class Label
A1	(2, 10)	C2
A2	(2, 6)	C1

A3	(11, 11)	C3
A4	(6, 9)	C2
A5	(6, 5)	C1
A6	(1, 2)	C1
A7	(5, 10)	C2
A8	(4, 9)	C2

Suppose we have a point $P=(5,7)$ for which we want to determine the class label using the K-nearest neighbor (KNN) algorithm. Which class does the P belongs?

- (A) C1
(B) C2
(C) C3
(D) Can't Predict

- Q11** Suppose you have a dataset with the following points in a two-dimensional space:

$\{(2,3), (3,5), (5,2), (8,8), (10,7), (12,10)\}$

Each point belongs to either Class A or Class B.

The classes are labeled as follows:

- Class A: $(2,3), (3,5), (5,2)$
- Class B: $(8,8), (10,7), (12,10)$

Now, a new point $(7,6)$ is given. Apply the k-Nearest Neighbors algorithm with $k=3$ to predict the class of this new point.

- (A) Class A
(B) Class B
(C) Can't Decide
(D) Both Class A and B

- Q12** In a KNN classifier with $K=5$, if the majority class among the 5 nearest neighbors is class A, what is the predicted class for a new data point?

- (A) Class A
(B) Class B
(C) Cannot be determined
(D) Depends on the distance metric

- Q13** Match List-I with List-II and select the correct answer using the codes given below the lists

List-I	List-II
A. K-Nearest Neighbors (KNN)	1. Discriminative Model
B. Forward selection	2. Dimensionality Reduction
C. Naive Bayes classifier	3. Generative Model

Code



- (A) A B C
1 2 3
- (B) A B C
1 3 2
- (C) A B C
2 3 1
- (D) A B C
2 1 3

Q14 Which of the following statements is/are true?

- (A) In k-Nearest Neighbors (k-NN), the classification is based on the majority vote among the k nearest neighbors.
- (B) Lasso Regression includes all features in the final model, regardless of their relevance.
- (C) Principal Component Analysis (PCA) uses Euclidean distance to perform dimensionality reduction.
- (D) Logistic Regression estimates the probability of a binary outcome using a linear function.

Q15 Which of the following statements are true?

- (A) Training a k-nearest-neighbors classifier takes more computational time than applying it.
- (B) The more training examples, the more accurate the prediction of a k-nearest-neighbors.
- (C) k-nearest-neighbors cannot be used for regression.
- (D) A k-nearest-neighbors is sensitive to outliers.

Q16 Consider these training examples:

x	y
(0,0)	1
(1,2)	1
(2,3)	0
(5,3)	0

How does 3-NN (three nearest neighbors) classify (2,1) using L1 distance?

Q17 Imagine you are using a k-Nearest Neighbor classifier on a dataset with lots of noise. You want your classifier to be less sensitive to the noise. Which of the following is likely to help and with what side effect?

- (A) Increase the value of $k \rightarrow$ Increase in prediction time
- (B) Decrease the value of $k \rightarrow$ Increase in prediction time
- (C) Increase the value of $k \rightarrow$ Decrease in prediction time
- (D) Decrease the value of $k \rightarrow$ Decrease in prediction time

Q18 A classifier maps a feature vector x in a vector space X to a discrete label y . Two classifiers C_1 and C_2 are considered equal if, for all $x \in X$, $C_1(x) = C_2(x)$, i.e., the output labels produced by both classifiers are identical for every possible input in the feature space.

Suppose we have a training dataset S and two 1-Nearest Neighbor (1-NN) classifiers, C_1 and C_2 , trained on S .

The equality of the classifiers is analyzed in the following cases: owing cases:

- Case (a): C_1 uses the L1-distance, and C_2 uses the L2-distance.
- Case (b): C_1 uses the L2-distance, and C_2 uses the square of the L2-distance.

Are C_1 and C_2 equal in the two cases described above?

- (A) C_1 is equal to C_2 in both case (a) and case (b).
- (B) C_1 is not equal to C_2 in case (a), but C_1 is equal to C_2 in case (b).
- (C) C_1 is equal to C_2 in case (a), but C_1 is not equal to C_2 in case (b).
- (D) C_1 is not equal to C_2 in either case (a) or case (b).



Answer Key

Q1 1~1
Q2 0~0
Q3 5~5
Q4 (A, B)
Q5 0~0
Q6 (B, C, D)
Q7 (A)
Q8 0~0
Q9 (A, B, C, D)

Q10 (C)
Q11 (A)
Q12 (A)
Q13 (A)
Q14 (A)
Q15 (B, D)
Q16 1~1
Q17 (A)
Q18 (B)



Hints & Solutions

Q1 Text Solution:

$$((1, 1), -1) \rightarrow 2 + 5 = 7$$

$$((1, 7), +1) \rightarrow 2 + 1 = 3$$

$$((3, 3), +1) \rightarrow 0 + 3 = 3$$

$$((5, 4), -1) \rightarrow 2 + 2 = 4$$

$$((2, 5), -1) \rightarrow 1 + 1 = 2$$

Test point (3, 6), now we find manhattan distance from each training point.

So, min distance is for 2, 3, 5, $\rightarrow 1, +1, -1, \Rightarrow +1$

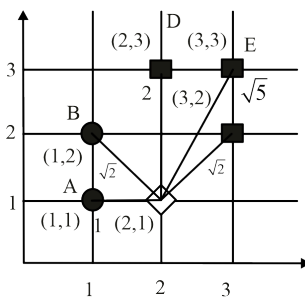
Q2 Text Solution:

	IMDB Rating	Duration	Genre	Test point 7.4, 114
1	8.0 (Mission Impossible)	160	Action	$\sqrt{.6^2 + 46^2}$
2	6.2(Gadar 2)	170	Action	$\sqrt{.2^2 + 56^2}$
3	7.2 (Rocky & Rani)	168	Comedy	$\sqrt{.2^2 + 54^2}$
4	8.2 (OMG 2)	155	Comedy	$\sqrt{.8^2 + 41^2}$

Test point 7.4, 114

So, 4th point is closet $\rightarrow 1NN \ y_{Barble} = y_{omg2}$
= comedy.

Q3 Text Solution:



- 1NN $\rightarrow A \rightarrow \bullet$
 - 3NN $\rightarrow A, B, C \rightarrow \bullet$
 - 5NN $\rightarrow ABCDE \rightarrow \bullet$
- K = 5

Q4 Text Solution:

A. K-NN can be used for both classification and regression.

- **True.** k-NN is a versatile algorithm that can be used for both **classification** (by majority

voting) and **regression** (by averaging the target values of k-nearest neighbors).

B. As k increases, the bias usually increases.

- **True.** When k is small, the model captures finer details (low bias, high variance). As k increases, the model generalizes more, leading to **higher bias** and a smoother decision boundary.

Q5 Text Solution:

Slow	Rarely	No
1	1	0 $\rightarrow$ 2
1	1	1 $\rightarrow$ 3
0	1	0 $\rightarrow$ 1
1	1	0 $\rightarrow$ 2
1	1	0 $\rightarrow$ 2
1	1	0 $\rightarrow$ 2
1	0	0 $\rightarrow$ 1
0	1	1 $\rightarrow$ 2
0	1	0 $\rightarrow$ 1
0	1	1 $\rightarrow$ 2
1	1	0 $\rightarrow$ 2
1	1	0 $\rightarrow$ 2

3NN $\rightarrow$ Find clam of test point

dimension are categorical in nature $\rightarrow$ Hamming distance

Hamming distance $\rightarrow$ dimension equal distance
=0

dimension not distance =1

Point 2,7,9 are NN $\rightarrow$ Animal

Animal

fish

So, Animal will be assigned.

Q6 Text Solution:

so weighted voting close neighbour are given more importance
outlier effect reduce overfit reduce Bias
increases

Q7 Text Solution:

- α_1 distance: manhattan distance
- α_2 distance: Euclidean distance



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- Training error@

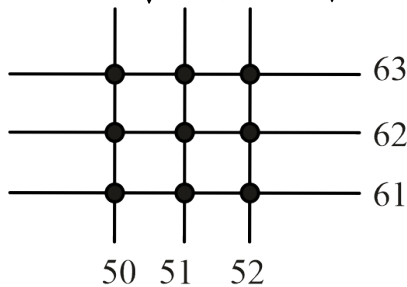
1NN $\rightarrow$ 0

Q8 Text Solution:

2D data 51.3, 62.1

Closest NN can be

- 51, 62, $\rightarrow \sqrt{.3^2 + .12} = \sqrt{.1}$
- 51, 63, $\rightarrow \sqrt{.3^2 + .9^2} = \sqrt{.9}$
- 52, 62 $\rightarrow \sqrt{.7^2 + .1^2} = \sqrt{.5}$
- 52, 63 $\rightarrow \sqrt{.7^2 + .9^2} = \sqrt{1.3}$



So, (51, 62), (51, 63), (52, 62) are

$\downarrow$ $\downarrow$ $\downarrow$
 (-1) odd (-1) even (-1) even
 (-1) 1 1 $\rightarrow$ Class 1.

Q9 Text Solution:

A. In case of a very large value of k, we may include points from other classes into the neighborhood.

- **True.** A very large k means the decision is influenced by distant points, which might belong to different classes, leading to misclassification.

B. In case of too small value of k, the algorithm is very sensitive to noise.

- **True.** With a small k (e.g., k=1), the prediction depends only on the closest point, making the algorithm highly sensitive to noise and outliers.

C. K-NN is a non-parametric algorithm.

- **True.** k-NN makes no assumptions about the underlying data distribution, meaning it is **non-parametric**. It determines the decision boundary based on the dataset rather than a predefined model.

D. k-NN algorithm does more computation on test time rather than train time.

- **True.** k-NN does **no explicit training**—it just stores the data. However, at test time, it must compute distances between the test point and all training points, making it computationally expensive.

Final Answer: A, B, C, and D are all true

Q10 Text Solution:

Point	Coordinates	Class Label	Test Point 5,7
A1	(2, 10)	C2	$\sqrt{3^2 + 3^2}$
A2	(2, 6)	C2	$\sqrt{3^2 + 1^2}$
A3	(11, 11)	C3	$\sqrt{6^2 + 4^2}$
A4	(6, 9)	C2	$\sqrt{12^2 + 2^2}$
A5	(6, 5)	C1	$\sqrt{12^2 + 2^2}$
A6	(1, 5)	C1	$\sqrt{4^2 + 5^2}$
A7	(5, 10)	C2	$\sqrt{3^2}$
A8	(4, 9)	C2	$\sqrt{1^2 + 2^2}$

So, 3NN $\rightarrow$ A4, A5, A8 $\Rightarrow$ C2

C₂ C₁ C₂

Q11 Text Solution:

- Class A: (2,3), (3,5), (5,2)
- Class B: (8,8), (10, 7), (12,10)

Distance from (7, 6)

$\downarrow$
 3NN $\rightarrow$ A,B,B

$\downarrow$
 B

$$2, 3 \rightarrow \sqrt{5^2 + 3^2} = \sqrt{34}$$

$$A \quad 3, 5 \rightarrow \sqrt{4^2 + 1^2} = \sqrt{17}$$

$$5, 2 \rightarrow \sqrt{2^2 + 4^2} = \sqrt{20}$$

$$B \quad 8, 8 \rightarrow \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$B \quad 10, 7 \rightarrow \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$12, 10 \rightarrow \sqrt{5^2 + 4^2} = \sqrt{41}$$

Q12 Text Solution:

In KNN classification, the predicted class for a new data point is determined by the majority class among its A nearest neighbors. Since the majority class among the 5 nearest neighbors is class A. the predicted class for the new data point is class A.



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Q13 Text Solution:

K-Nearest Neighbors (KNN) → Discriminative Model
 Forward selection → Dimensionality Reduction
 Naive Bayes classifier → Generative Model

Q14 Text Solution:

- k-NN classifies based on the majority vote among its k nearest neighbors. So, option (a) is correct.
- Lasso regression can set some coefficients to zero, effectively excluding less relevant features. So, option (b) is incorrect.
- PCA performs dimensionality reduction using orthogonal transformations, not Euclidean distance. So, option (c) is incorrect.
- Logistic regression uses the logistic function (sigmoid) to estimate probabilities, not a linear function. So, option (d) is incorrect.

Q15 Text Solution:

B. The more training examples, the more accurate the prediction of a k-nearest-neighbors.

- **True.** More training data generally improves accuracy, as k-NN relies on having enough representative neighbors to make better decisions. However, after a certain point, adding more data may not significantly improve performance and increases computational cost.

D. A k-nearest-neighbors is sensitive to outliers.

- **True.** Since k-NN makes predictions based on local neighbors, **outliers** (which are far from the main distribution) can **mislead predictions**, especially when k is small.

Q16 Text Solution:

x	y	2,1
(0,0)	1	→ $2 + 1 = 3$
(1,2)	1	→ $1 + 1 = 2$
(2,3)	0	→ $0 + 2 = 2$
(5,3)	0	→ $3 + 2 = 5$

Q17 Text Solution:

A. Increase the value of k → Increase in prediction time

- **True.** Increasing k makes the model **less sensitive to noise** because it averages over more neighbors, reducing the impact of individual noisy points.
- **Side effect:** Higher k means more neighbors need to be considered, **increasing prediction time** as more distance calculations are required.

Q18 Text Solution:

Case b we know if $\sqrt{a}$ is min then a is also min.

Ex.

A(0, 0)

B(1, 1)

C(1.5, 5)

Test point (1, 0)

$$\begin{array}{l}
 \alpha 1 \text{ distance } A \rightarrow d \\
 B \xrightarrow{\quad \quad \quad} 1 \\
 C \rightarrow .5 + .5 = 1 \\
 \alpha 2 \text{ distance } A = \sqrt{1^2} = 1 \\
 B = \sqrt{1^2} = 1 \\
 C = \sqrt{.5^2 + .5^2} = \sqrt{.5}
 \end{array}
 \left. \vphantom{\begin{array}{l} \alpha 1 \text{ distance } A \rightarrow d \\ B \xrightarrow{\quad \quad \quad} 1 \\ C \rightarrow .5 + .5 = 1 \end{array}} \right\} \text{all 3 equal}$$



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